Nirbhay Singh

GCP Cloud Architect & DevOps Engineer | 5x Google Cloud Certified

5x	11+	EUR 4M+	15+	14+
GCP Certified	Years Experience	Cloud Savings	Business Units	Total Certs

PROFESSIONAL SUMMARY

5x Google Cloud certified Cloud & DevOps Architect with 11+ years of production experience. Deep expertise in GKE, Vertex AI, BigQuery, Cloud Run, Cloud Build, and Anthos for enterprise-scale platform engineering. Currently leading the strategic Google Cloud partnership at Bosch Europe -- designing GKE platforms, Vertex AI pipelines, and BigQuery analytics adopted across 15+ business units. Proven ability to architect scalable, secure GCP infrastructure with FinOps governance, delivering EUR 4M+ in documented cloud savings.

KEY ACHIEVEMENTS

- 5x Google Cloud Certified: Professional ML Engineer, Cloud DevOps Engineer, Cloud Architect, Generative AI Leader, Associate Cloud Engineer
- Leading strategic Google Cloud partnership for Bosch Poland -- GCP architecture standards, governance, and platform engineering
- Designed Vertex AI and GKE-based ML/AI platforms adopted across 15+ Bosch Europe business units for enterprise search and automation
- Architected production GKE clusters with Anthos service mesh, Binary Authorization, and Workload Identity for zero-trust security
- Built BigQuery analytics pipelines processing petabyte-scale enterprise data with optimised partitioning and materialised views
- Deployed Cloud Run microservices for serverless AI inference with auto-scaling, traffic splitting, and canary deployments
- Implemented GCP FinOps: committed use discounts, preemptible VMs, BigQuery slot reservations, contributing to EUR 4M+ savings

PROFESSIONAL EXPERIENCE

Cloud & AI Architect (GCP Lead)

Dec 2022 - Present

Robert Bosch (Bosch Polska) Warsaw, Poland

- Led strategic Google Cloud partnership for Bosch Poland -- defined GCP landing zones, architecture standards, and governance policies
- Designed Vertex AI and BigQuery-powered ML/AI solutions adopted across 15+ Bosch Europe business units for enterprise search and process automation
- Architected production GKE platforms with Anthos service mesh, Workload Identity, Binary Authorization, and namespace-level multi-tenancy
- Built BigQuery analytics pipelines with optimised partitioning, clustering, and materialised views for petabyte-scale data processing
- Deployed Cloud Run microservices for serverless LLM inference with traffic splitting, auto-scaling, and canary rollouts
- Implemented GCP FinOps governance: committed use discounts, preemptible VMs, slot reservations, contributing to EUR 4M+ total savings
- Mentored teams on GCP Well-Architected Framework, GKE operations, Vertex AI best practices, and Cloud Build pipeline design

Senior DevOps Consultant

Jun 2021 - Nov 2022

McAfee

- Deployed and managed production GKE clusters with node auto-provisioning, pod auto-scaling, and automated security patching
- Built CI/CD pipelines using Cloud Build and GitHub Actions for GKE deployments with automated testing and canary analysis
- Automated GCP infrastructure with Terraform (GCP provider), achieving 50%+ reduction in deployment time
- Implemented comprehensive observability with Cloud Monitoring, Cloud Logging, Prometheus, and Grafana dashboards

Nirbhay Singh

GCP | GKE | Vertex AI | BigQuery
Cloud Build | Anthos | Terraform

CONTACT

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CORE SKILLS

GCP COMPUTE

GKE, Cloud Run, Compute Engine, Cloud Functions, App Engine, Anthos

GCP AI/ML

Vertex AI, AutoML, GenAI Studio, BigQuery ML, AI Platform Pipelines

GCP DATA

BigQuery, Cloud Storage, Dataflow, Dataproc, Pub/Sub, Cloud SQL, Firestore

GCP SECURITY

IAM, Cloud KMS, SCC, Binary Auth, VPC SC, Workload Identity

GCP DEVOPS

Cloud Build, Cloud Deploy, Artifact Registry, Config Connector

GCP NETWORKING

VPC, CLB, Cloud CDN, Cloud DNS, Interconnect, Private Google Access

IAC & CI/CD

Terraform (GCP), GitHub Actions, Jenkins, ArgoCD, GitOps, Flux

OBSERVABILITY

Cloud Monitoring, Cloud Logging, Cloud Trace, Prometheus, Grafana

EDUCATION

MTech (Master of Technology)

Birla Institute of Technology

LANGUAGES

English	Professional
Hindi	Native

Senior DevOps Engineer / Data Architect

Dec 2018 - Jun 2021

Deloitte

- Built BigQuery data platforms for enterprise analytics with automated ETL using Dataflow and Cloud Composer
- Implemented multi-cloud security controls spanning GCP, AWS, Azure, and OCI with unified audit logging and IAM
- Orchestrated containerised workloads across GKE and EKS for distributed data processing at scale
- Accelerated CI/CD pipeline speed by 30x through Cloud Build optimisation, caching, and parallel execution

Senior Cloud Infrastructure Engineer

Nov 2015 - Dec 2018

PTC

- Designed cloud architectures for production workloads -- established patterns later applied to GCP migrations
- Built disaster recovery strategies with cross-region replication and automated failover testing

System Engineer

Mar 2013 - Jul 2014

IBM

- Infrastructure operations, P1/P2 incident response, and root cause analysis for enterprise production systems

OPEN-SOURCE PROJECTS

InfraCents -- GCP Cost Estimates on Every Terraform PR

>> github.com/nirbhays/infracents

GitHub App posting real-time GCP cost estimates on Terraform pull requests. Queries live GCP Pricing API, parses Terraform changes automatically. Zero config.

STACK: Python, Next.js, Terraform, GitHub App, GCP Pricing API

ShieldIaC -- AI-Powered IaC Security Scanner

>> github.com/nirbhays/shieldiac

Scans Terraform for GCP security misconfigurations pre-production. 100+ rules across CIS GCP Benchmark, SOC2, HIPAA, PCI-DSS, NIST. GPT-4.1 fix suggestions.

STACK: Python, Terraform, GPT-4.1, GitHub Actions

TokenMeter -- LLM Cost Intelligence (Google AI Support)

>> github.com/nirbhays/tokenmeter

Tracks token cost across Google AI (Gemini), OpenAI, Anthropic. Smart routing matches tasks to optimal models, saving up to 60% on AI spend.

STACK: Python, Google AI, OpenAI, Anthropic, Smart Routing, PyPI

CERTIFICATIONS & CREDENTIALS

- **Google** Professional Machine Learning Engineer (Valid Dec 2027)
- **Google** Professional Cloud Architect (Valid Jun 2026)
- **Google** Associate Cloud Engineer (Aug 2023)
- **HashiCorp** Terraform Associate
- **AWS** Solutions Architect - Associate (Jan 2021)
- **Google** Professional Cloud DevOps Engineer (Valid Dec 2027)
- **Google** Generative AI Leader (Valid Aug 2028)
- **CKAD** - Certified Kubernetes Application Developer (Valid Sep 2026)
- **AWS** Solutions Architect - Professional (Feb 2020)
- **MCSA** - Microsoft Certified Solutions Associate (Apr 2015)